

ANALYSTLAB AFRICA
Experience Lab Internship Programme



HealthConnect Clinic
Model Improvement & Validation Notebook/Report

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and Healthcare Support Using Data and AI
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This report is the Week 6 Data Science track deliverable for the HealthConnect Clinic Experience Lab engagement. It documents the Model Improvement & Validation work carried out in the notebook `wk6_Model_Improvement_Error_Analysis_Validation.ipynb`, which builds directly on the Week 5 baseline classification workflow rather than repeating it. The report is grounded exclusively in the cells, executed outputs and narrative decisions recorded in that notebook.

In Week 5, an interpretable Logistic Regression baseline model was developed to predict HealthConnect appointment no-shows, covering data preparation, exploratory analysis, feature engineering, leakage-aware preprocessing, model development and initial evaluation. The baseline established an initial performance benchmark for distinguishing between attended and no-show appointments using patient and appointment information available before the scheduled appointment. The key limitation carried into Week 6 was that the baseline represented only an initial predictive benchmark and required further investigation of its false-positive and false-negative errors, feature adequacy, and overall suitability for HealthConnect's intended use case, together with continued vigilance around data-leakage risk (particularly information not genuinely available at prediction time).

In Week 6, the baseline was therefore not rebuilt from scratch. Instead, the Week 5 processed training and test feature sets, the Week 5 target variables, and the exact saved Week 5 Logistic Regression model artefact were reloaded, and a series of validation assertions (matching feature columns, matching row counts, absence of missing values, and — where exposed by the fitted estimator — matching feature names and feature count) were run to confirm the loaded artefacts were fully compatible before any new analysis was performed. This reproducibility check is itself a methodological safeguard: it confirms that the Week 6 error analysis is being conducted against the genuine Week 5 baseline rather than an approximation of it.

The scope of Week 6 covered: analysing the weaknesses and errors of the Week 5 baseline; investigating false positives and false negatives at both a descriptive and a probability/confidence level; reviewing and refining features where the error analysis justified it; developing at least one improved modelling approach; comparing the improved model(s) fairly against the Logistic Regression baseline; incorporating relevant cross-track findings from the Data Analytics track; validating model performance and limitations; and recommending a candidate model for further testing in Week 7.

1. Baseline performance summary

The Week 5 Logistic Regression baseline was reproduced on the preserved Week 5 holdout test set (966 appointments: 483 Attended, 483 No-Show) using the saved model artefact, giving the following performance:

Metric	Value
Accuracy	0.6366
Precision (No-Show)	0.6304
Recall (No-Show)	0.6605
F1-Score (No-Show)	0.6451
ROC-AUC	0.6873
PR-AUC	0.6680

The corresponding confusion matrix on the test set was:

Outcome	Count
True Negatives (correctly predicted Attended)	296
False Positives (predicted No-Show, actually Attended)	187
False Negatives (predicted Attended, actually No-Show)	164
True Positives (correctly predicted No-Show)	319

These figures reproduce, exactly, the Week 5 baseline as recorded in the Week 5 Baseline Modelling Report, confirming that no drift was introduced by reloading the saved artefacts. From this baseline, three weaknesses were carried forward into the Week 6 error analysis: (i) a broadly balanced but still substantial volume of both False Positives (187) and False Negatives (164), (ii) moderate discriminatory power (ROC-AUC 0.6873, PR-AUC 0.6680) that leaves meaningful room for improvement, and (iii) a No-Show recall of only 66.05%, meaning roughly a third of genuine no-shows were not being identified by the baseline at its default 0.50 classification threshold.

2. Error analysis

An error-analysis dataset was constructed by attaching the actual outcome, the predicted class, the predicted no-show probability, and a derived `error_type` label (True Negative / False Positive / False Negative / True Positive) to every row of the Week 5 test set. This enabled row-level investigation of where and why the baseline model was wrong, rather than relying on aggregate metrics alone.

2.1 Weaknesses Identified from the Confusion Matrix

- Balanced but substantial error types: the baseline produces nearly as many False Positives (187) as False Negatives (164), meaning it is almost equally likely to wrongly predict a no-show that does not occur as to miss one that does.
- False Negatives (missed no-shows) are the more operationally costly error: 164 missed no-shows mean the clinic cannot intervene (reminders, rescheduling, overbooking) for these patients, directly wasting appointment slots and revenue.
- False Positives (false alarms) are less costly but not free: 187 patients who would have attended risk unnecessary reminders, overbooking, or a perception of being unfairly flagged.
- No-Show recall of 66.05% means roughly one in three genuine no-shows is missed at the default threshold — a key area for improvement if the objective is proactive no-show management.

2.2 Investigating False Negatives

False Negatives (actual No-Show, predicted Attended; $n = 164$) were compared descriptively against True Positives. False Negatives showed, on average, shorter booking lead times (standardised mean -0.62 , versus $+0.52$ for False Positives), fewer previous no-shows (standardised mean -0.30), a high rate of reminders having been sent (79.3%), and a somewhat higher share of morning appointments (51.2% of False Negatives were morning appointments).

Distributional plots of age, booking lead time, previous appointments and distance to clinic, compared between False Negatives and True Positives, reinforced that False Negatives cluster among patients whose observable characteristics would typically be read as lower-risk — shorter lead time, a cleaner attendance history, and closer or unremarkable distance to the clinic.

Confidence analysis of the ten lowest-probability False Negatives showed predicted no-show probabilities of roughly 0.19–0.27 — well below the 0.50 threshold — meaning the model was not merely uncertain about these patients but confidently wrong: its risk signal pointed clearly toward attendance for patients who, in fact, did not show.

2.3 Investigating False Positives

False Positives (actual Attended, predicted No-Show; $n = 187$) showed the mirror-image pattern: longer booking lead times (standardised mean +0.52), a higher previous no-show history (standardised mean +0.23), a broadly similar rate of reminders sent (73.8%), and a slightly higher share of evening appointments (18.2%) than False Negatives. Confidence analysis of the ten highest-probability False Positives showed predicted no-show probabilities of roughly 0.80–0.87, again indicating confidently incorrect predictions rather than borderline, low-confidence calls.

2.4 Summary of Error Analysis

- The model is confidently wrong in both directions, not merely uncertain: the highest-probability False Positives and the lowest-probability False Negatives both sit well away from the 0.50 decision boundary.
- `booking_lead_days` and `previous_no_shows` drive both error types, but in opposite directions — shorter lead time / lower prior no-show history is associated with missed no-shows (False Negatives), while longer lead time / higher prior no-show history is associated with false alarms (False Positives). This indicates the baseline is not capturing the nuance or interaction between these two features.
- Reminders being sent does not reliably prevent a no-show in the False Negative group, suggesting the model may over-weight `reminder_sent` as a positive signal for attendance for some patient segments.
- A review of permutation importance against these findings (Section 6.2) confirmed that the dominant baseline features — `booking_lead_days` and `previous_no_shows` — are directionally correct but that their interactions and non-linear effects were not represented in the original feature set, motivating the feature refinement described in Section 3.

3. Feature refinement

The error analysis indicated that the baseline's two dominant features — `booking_lead_days` and `previous_no_shows` — behaved differently for False Negatives than for False Positives, suggesting the linear baseline was not capturing how these features interact with one another or with distance. Three interaction features were engineered as testable hypotheses, added to copies of the Week 5 processed training and test sets (`X_train_refined`, `X_test_refined`) so that the original Week 5 artefacts remained unchanged:

New Feature	Definition	Rationale
history_x_lead_time	previous_no_shows × booking_lead_days	Tests whether a history of no-shows compounds with a long booking lead time to produce a particularly high-risk profile.
distance_x_lead_time	distance_to_clinic_km × booking_lead_days	Tests whether distance becomes more predictive of no-show when an appointment is booked well in advance, as circumstances have more time to change.
history_interaction	previous_appointments × previous_no_shows	Captures a more holistic view of attendance reliability, distinguishing patients with many appointments and few no-shows from those with a similar volume but a higher no-show proportion.

This raised the feature count from 35 (Week 5 baseline) to 38 (Week 6 refined set). The notebook explicitly frames these additions as hypotheses requiring validation through retraining and evaluation rather than assumed improvements — a discipline that is reflected in Section 5, where the refined feature set is compared fairly against the unmodified baseline features.

4. Improved model(s)

Three candidate improvements were trained on the refined feature set (`X_train_refined`) and evaluated on the corresponding refined test set (`X_test_refined`), alongside the unmodified Week 5 baseline for comparison:

- **Refined Logistic Regression** — the same linear model family as the baseline, retrained on the refined 38-feature set with `class_weight="balanced"` and the liblinear solver, isolating the effect of feature refinement from any change in model family.
- **Random Forest Classifier** — 400 trees, `min_samples_leaf=4`, `class_weight="balanced"`, introduced as a non-linear alternative capable of capturing feature interactions and non-linear decision boundaries without manual engineering.
- **Histogram Gradient Boosting Classifier** — `learning_rate=0.05`, `max_iter=200`, `max_leaf_nodes=15`, `l2_regularization=1.0`, introduced as a boosting alternative that sequentially corrects the errors of prior trees.

All three models were trained with a fixed `random_state=42` for reproducibility and evaluated using the same metric suite as the baseline (Accuracy, Precision, Recall, F1, ROC-AUC, PR-AUC), so that the comparison in Section 5 is on a like-for-like basis.

5. Model comparison

The four models were evaluated on the Week 5 holdout test set (or its refined-feature equivalent, as applicable):

Model	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Week 5 Logistic Regression (baseline)	0.6366	0.6304	0.6605	0.6451	0.6873	0.6680
Refined Logistic Regression	0.6325	0.6333	0.6294	0.6314	0.6890	0.6717
Random Forest	0.6284	0.6250	0.6418	0.6333	0.6637	0.6556
Histogram Gradient Boosting	0.6346	0.6255	0.6708	0.6474	0.6615	0.6537

The Refined Logistic Regression achieved the strongest ROC-AUC (0.6890) and PR-AUC (0.6717) of the four models, a modest but genuine improvement over the unmodified baseline (0.6873 / 0.6680), confirming that the engineered interaction features carried a small amount of additional, useful signal. The original baseline remained a close second and, notably, outperformed both tree-based models on ROC-AUC and PR-AUC. Random Forest and Histogram Gradient Boosting, despite their theoretical capacity to model non-linear relationships, underperformed both Logistic Regression variants with their initial/default hyperparameters — indicating that more model complexity did not translate into better generalised performance without further tuning.

5.1 Overfitting / Generalisation Analysis

Each model was additionally evaluated on its own training data to check for overfitting (a large train–test performance gap):

Model / Split	Accuracy	Precision	Recall	F1	ROC-AUC	PR-AUC
Logistic Regression (Train)	0.6356	0.6415	0.6613	0.6513	0.6894	0.6970
Logistic Regression (Test)	0.6366	0.6304	0.6605	0.6451	0.6873	0.6680
Refined Logistic Regression (Train)	0.6359	0.6504	0.6320	0.6410	0.6900	0.6977
Refined Logistic Regression (Test)	0.6325	0.6333	0.6294	0.6314	0.6890	0.6717
Random Forest (Train)	0.9459	0.9474	0.9474	0.9474	0.9903	0.9911
Random Forest (Test)	0.6284	0.6250	0.6418	0.6333	0.6637	0.6556
Histogram Gradient Boosting (Train)	0.7887	0.7978	0.7892	0.7935	0.8766	0.8905
Histogram Gradient Boosting (Test)	0.6346	0.6255	0.6708	0.6474	0.6615	0.6537

Both Logistic Regression models show a very small train–test gap in ROC-AUC (0.6894 → 0.6873 for the baseline; 0.6900 → 0.6890 for the refined model), indicating good generalisation and minimal overfitting. Random Forest, by contrast, shows a severe gap (Train ROC-AUC 0.9903 versus Test ROC-AUC 0.6637), evidencing heavy overfitting to the training data. Histogram Gradient Boosting shows a smaller but still substantial gap (0.8766 → 0.6615), also indicating meaningful overfitting. This is a critical finding: with their current, untuned

hyperparameters, neither tree-based model can be regarded as reliable, since their apparent training-set capability does not transfer to unseen data.

6. Evaluation results

6.1 Refined Logistic Regression Coefficients

Because the Refined Logistic Regression was identified as the leading candidate (Sections 5 and 9), its coefficients were examined directly to understand which features increase or decrease the predicted probability of a no-show. The fifteen largest-magnitude coefficients were:

Feature	Coefficient	Direction
booking_lead_days	+0.599	Increases no-show probability
appointment_day_of_year	+0.431	Increases no-show probability
appointment_month	-0.390	Decreases no-show probability
appointment_day_Wednesday	+0.302	Increases no-show probability
previous_no_shows	+0.293	Increases no-show probability
gender_Prefer not to say	-0.269	Decreases no-show probability
appointment_day_Thursday	+0.267	Increases no-show probability
appointment_day_Monday	+0.209	Increases no-show probability
booking_month	-0.196	Decreases no-show probability
appointment_type_General Consultation	-0.188	Decreases no-show probability
appointment_day_Saturday	+0.178	Increases no-show probability
distance_to_clinic_km	+0.166	Increases no-show probability
reminder_sent_Yes	-0.141	Decreases no-show probability
reminder_channel_WhatsApp	+0.137	Increases no-show probability
booking_week_of_year	+0.126	Increases no-show probability

booking_lead_days is the strongest positive predictor: longer lead times between booking and appointment are associated with a materially higher no-show probability, consistent with the intuition that appointments booked far in advance are more likely to be forgotten or superseded. previous_no_shows confirms that a patient's attendance history is directly predictive of future behaviour. distance_to_clinic_km and reminder_sent_Yes behave in the expected directions — greater distance increases risk, and a reminder having been sent decreases it. Several appointment-day and appointment-timing terms (Wednesday, Thursday, Monday, Saturday, appointment_month, appointment_day_of_year) also carry meaningful coefficients, indicating temporal patterns in attendance that were not fully captured by the baseline's simpler feature encoding.

6.2 Permutation Feature Importance

Permutation importance (ROC-AUC scoring, 20 repeats) was computed for the Refined Logistic Regression to provide a model-agnostic view of which features actually drive predictive performance, independent of coefficient scaling:

Feature	Mean Importance (Δ ROC-AUC)
booking_lead_days	0.1594
appointment_month	0.0432
appointment_day_of_year	0.0401
previous_no_shows	0.0211
booking_month	0.0188
distance_to_clinic_km	0.0062
appointment_day_Monday	0.0054
booking_day_of_week	0.0032
age	0.0028

booking_lead_days is by a wide margin the single most important feature — shuffling it alone reduces ROC-AUC by 0.159, roughly four times the next-largest effect. Temporal features (appointment_month, appointment_day_of_year, booking_month) and previous_no_shows form a clear second tier of importance, while distance, age and day-of-week effects contribute more modestly. The engineered interaction terms (history_x_lead_time, distance_x_lead_time, history_interaction) appeared in the wider top-15 permutation ranking but with small individual importance values, indicating the model is using them, but only to a limited extent — consistent with the modest (not transformative) performance gain observed in Section 5.

6.3 Leakage-Safe Threshold Optimisation

The Refined Logistic Regression was taken forward for threshold analysis as the strongest candidate on ROC-AUC/PR-AUC. To avoid tailoring the operating threshold to the final test set, threshold selection was performed entirely on out-of-fold (OOF) probabilities generated from the training data via five-fold stratified cross-validation (cross_val_predict, StratifiedKFold, n_splits=5, shuffle=True, random_state=42). The processed Week 5 feature matrices do not retain patient identifiers, so this cross-validation could not reproduce the patient-grouped separation used in the original Week 5 train/test split; this limitation is carried forward as an explicit Week 7 requirement (Section 12.2).

Because HealthConnect's objective is to reduce missed appointments while improving, not degrading, the patient support experience, the operating threshold was not selected by simply maximising F1 or F2. Instead, nine candidate thresholds (0.30 to 0.70, in steps of 0.05) were each assessed against the default 0.50 threshold on the OOF predictions using four business-aligned measures — Recall Gain, False-Negative (FN) Reduction, Precision Change, and Additional False Positives (FP) — together with a derived intervention-efficiency measure, Extra FP per FN Avoided (the number of additional unnecessary patient contacts generated for every additional missed no-show that is prevented, relative to the 0.50 threshold):

Threshold	Precision	Recall	F1	Recall Gain	FN Reduction	Additional FP	Extra FP / FN Avoided
0.30	0.5532	0.9294	0.6936	+0.3036	589	769	1.306
0.35	0.5751	0.8706	0.6926	+0.2448	475	561	1.181
0.40	0.5938	0.8031	0.6827	+0.1773	344	379	1.102
0.45	0.6157	0.7216	0.6645	+0.0959	186	187	1.005
0.50 (default)	0.6386	0.6258	0.6321	0.0000	0	0	—
0.55	0.6585	0.5268	0.5853	-0.0990	-192	-157	—
0.60	0.6977	0.4139	0.5196	-0.2119	-411	-339	—
0.65	0.7291	0.3052	0.4302	-0.3206	-622	-467	—
0.70	0.7773	0.2052	0.3246	-0.4206	-816	-573	—

The provisional operating threshold was then selected by an explicit, pre-specified rule: among candidate thresholds below 0.50 that simultaneously increase Recall, reduce False Negatives, and do not reduce False Positives below the 0.50 baseline, the threshold with the minimum Extra FP per FN Avoided was chosen as the most intervention-efficient option. Applying this rule to the OOF results above selected a threshold of 0.45 (Extra FP per FN Avoided = 1.005 — the lowest among the four qualifying candidates), in preference to the statistically F1-/F2-optimal threshold of 0.30, which was rejected as operationally inefficient (Extra FP per FN Avoided = 1.306, more than 25% higher). Because HealthConnect has not supplied a formal tolerance for unnecessary outreach, this selected threshold (SELECTED_THRESHOLD = 0.45) is explicitly treated as provisional pending Week 7 stakeholder validation (Section 12.1), not as a finalised operating point.

The frozen 0.45 threshold — selected using only training-data OOF evidence — was then evaluated once, for the first time, on the untouched Week 5 holdout test set, alongside the default 0.50 threshold, to test whether the training-validation trade-off generalises to unseen patients:

Configuration	Threshold	Accuracy	Precision	Recall	F1	FP	FN
Refined LR — default threshold	0.50	0.6325	0.6333	0.6294	0.6314	176	179
Refined LR — HealthConnect-aligned threshold	0.45	0.6263	0.6066	0.7184	0.6578	225	136

On the holdout test set, moving from the default 0.50 threshold to the HealthConnect-aligned 0.45 threshold produced a recall gain of +0.0890 (62.94% → 71.84%) and avoided 43 false negatives (missed no-shows), at the cost of a precision decline of -0.0267 and 49 additional false positives — an efficiency ratio of 1.140 extra false-positive contacts per false negative avoided on the holdout set, close to the 1.005 observed during training-only validation. ROC-AUC (0.6890) and PR-AUC (0.6717) are unchanged across both configurations, since adjusting the classification threshold changes only the decision rule, not the underlying probability ranking. This close agreement between the training-only selection evidence and the independent holdout result is a meaningful (though not conclusive) sign that the trade-off is not

an artefact of the specific training fold split, though full confirmation requires the robust cross-validation specified for Week 7 (Section 12.2).

7. Interpretation

Read together, the coefficient analysis, permutation importance and error analysis tell a consistent story. `booking_lead_days` and `previous_no_shows` are, and remain, the dominant genuine risk signals for HealthConnect no-shows — both the Data Analytics track's descriptive findings (Section 11) and the model's own internal importance measures agree on this. However, the baseline's difficulty was never in identifying that these features matter; it was in correctly weighting them for patients whose overall profile looked deceptively safe (short lead time, clean history) or deceptively risky (long lead time, prior no-shows) at the margins. The interaction terms introduced in Week 6 partially address this — the small ROC-AUC and PR-AUC gains achieved by the Refined Logistic Regression (+0.0017 ROC-AUC, +0.0037 PR-AUC over baseline) show that some of this nuance was captured, but the gain is modest rather than transformative.

The overfitting analysis (Section 5.1) is equally important to the interpretation: it shows that the apparent capacity of Random Forest and Histogram Gradient Boosting to model non-linear interactions did not, in Week 6, translate into better real-world (test-set) performance. Both tree-based models learned the training data — including its noise — too well, and their untuned test performance was worse than either Logistic Regression variant. This means the correct reading of Week 6 is not "tree-based models failed to help" but "tree-based models require proper hyperparameter tuning and regularisation before their potential can be assessed fairly" — a task explicitly deferred to future iterations rather than attempted in Week 6.

The threshold-optimisation results (Section 6.3) add a further, business-relevant layer of interpretation. Because the Refined Logistic Regression's default-threshold performance is only marginally better than the baseline's, the more operationally meaningful lever, at this stage, is not model selection but threshold selection: moving from the default 0.50 threshold to the HealthConnect-aligned 0.45 threshold avoided 43 false negatives on the holdout test set at a cost of 49 additional false positives — a trade-off that could not previously be quantified in this report and is now grounded in leakage-safe, out-of-fold evidence rather than test-set-tuned figures. This resolves a methodological gap flagged in an earlier version of this report, in which several notebook narrative cells referenced threshold-sensitivity tables that had not yet been constructed by an executed code cell; those tables (`threshold_df_refined` and `threshold_business_df`) have since been built and are reported in full in Section 6.3.

This threshold evidence should nonetheless be read as provisional, not final. HealthConnect has not supplied a formal maximum acceptable false-positive contact rate, intervention-capacity limit, or patient-experience tolerance, so the 0.45 threshold was selected using an explicit, defensible efficiency rule (minimum Extra FP per FN Avoided among qualifying candidates) rather than a stakeholder-confirmed cost weighting. It is also, so far, validated only against ordinary stratified cross-validation rather than the patient-grouped separation used for the original Week 5 split, since the processed feature matrices do not retain patient identifiers. Both points are carried forward as explicit Week 7 requirements (Sections 12.1 and 12.2) rather than treated as settled.

8. Business relevance

HealthConnect's stated objective is to reduce missed appointments while improving, not degrading, the patient support experience. Evaluated against this objective, the Refined Logistic Regression model shows real but partial suitability:

- **Reducing missed appointments (default 0.50 threshold):** at the default threshold, the refined model identifies 62.9% of actual no-shows (304 of 483), leaving 179 False Negatives — missed opportunities to intervene with reminders, rescheduling or overbooking.
- **Reducing missed appointments (HealthConnect-aligned 0.45 threshold):** moving to the provisional 0.45 threshold selected in Section 6.3 raises recall to 71.8% (347 of 483 no-shows identified), avoiding 43 additional missed no-shows on the holdout test set relative to the default threshold, at the cost of 49 additional False Positives (225 versus 176) and a precision decline from 63.3% to 60.7%.
- **Improving patient support experience:** at the default threshold, 63.3% of appointments flagged as at-risk genuinely are (Precision = 0.633), meaning most triggered interventions are justified; at the HealthConnect-aligned 0.45 threshold this falls to 60.7%, meaning a modestly larger share of patients who would in fact attend could be subjected to unnecessary reminders or a perceived sense of being monitored.
- **The Precision–Recall trade-off is now quantified, not just described:** the leakage-safe threshold-optimisation work in Section 6.3 shows that lowering the threshold from 0.50 to 0.45 costs approximately 1.14 additional unnecessary patient contacts for every additional missed no-show avoided (Extra FP per FN Avoided) on the holdout test set — a concrete, business-interpretable figure that HealthConnect stakeholders can weigh against their own tolerance for unnecessary outreach in Week 7.
- **Cost asymmetry should still drive the final operating point:** because a missed no-show (False Negative) typically costs the clinic a wasted appointment slot and lost revenue, while a false alarm (False Positive) costs at most an unnecessary reminder or contact, the 0.45 threshold is offered as a provisional, evidence-based starting point rather than a final recommendation — HealthConnect stakeholders should confirm whether an intervention-efficiency ratio of roughly 1.14 extra contacts per no-show avoided is operationally acceptable.

In short, the Refined Logistic Regression model, together with the HealthConnect-aligned 0.45 threshold, is a genuine, interpretable, and now business-quantified improvement over the Week 5 default-threshold baseline, and is ready to progress to Week 7 testing. It is not yet, on its own, a finished solution for reducing HealthConnect no-shows — its business value will be realised through stakeholder confirmation of the operating threshold, human-in-the-loop intervention design, and (per Section 10) fairness testing before any production deployment.

9. Candidate model recommendation

The Refined Logistic Regression model, evaluated at its HealthConnect-aligned provisional threshold of 0.45, is retained as the provisional candidate for progression into Week 7 testing — provisional because its performance should be treated as a decision-support capability rather than evidence of deployment readiness, and because both the small headline improvement over the baseline and the selected threshold still require further validation. Its performance on the holdout test set, at both the default and the HealthConnect-aligned threshold, is:

Metric	Default (0.50)	HealthConnect-aligned (0.45)
ROC-AUC	0.6890	0.6890
PR-AUC	0.6717	0.6717
Precision (No-Show)	0.633	0.607
Recall (No-Show)	0.629	0.718
F1-Score (No-Show)	0.631	0.658
False Negatives (missed no-shows)	179	136
False Positives (false alarms)	176	225

The recommendation is based on the following reasoning:

1. Strongest numerical ranking metrics among the models evaluated with initial/default parameters — highest ROC-AUC and PR-AUC of the four candidates compared in Section 5, though the margin over the baseline (+0.0017 ROC-AUC, +0.0037 PR-AUC) is small and not yet shown to be stable across patient-grouped validation.
2. Excellent generalisation: the smallest train–test performance gap of any model tested, indicating a low risk of the model having overfit to the training data (Section 5.1).
3. Interpretability: as a Logistic Regression model, its coefficients (Section 6.1) can be explained directly to HealthConnect stakeholders, supporting trust and stakeholder buy-in ahead of any production use.
4. Effective, evidence-based feature engineering: the three interaction terms were motivated directly by the Week 6 error analysis (Section 2) and Data Analytics cross-track findings (Section 11), and produced a genuine, if modest, performance improvement over the unmodified baseline.
5. HealthConnect-aligned, leakage-safe threshold validation: the provisional 0.45 operating threshold was selected using out-of-fold training evidence only (Section 6.3) and, when evaluated once on the untouched holdout test set, avoided 43 additional false negatives at a cost of 49 additional false positives (1.14 extra contacts per no-show avoided) — providing a concrete, quantified basis for HealthConnect stakeholders to confirm or adjust the operating point in Week 7.

Random Forest and Histogram Gradient Boosting are not recommended for progression in their current, untuned form, given their demonstrated overfitting (Section 5.1) and weaker test-set performance than either Logistic Regression variant. They remain candidates worth revisiting only if properly regularised and hyperparameter-tuned in a future iteration.

10. Limitations

10.1 Limitations of the Refined Logistic Regression Model

- Linearity assumption: as a linear model, even with three added interaction terms, it may still not capture highly complex, non-linear relationships between features and no-show behaviour that a properly tuned tree-based model could.

- **Feature completeness:** the model relies solely on the features available in the current dataset. Other potentially important factors (e.g. socio-economic status, specific medical conditions, transportation access) are not present and could be limiting the model's predictive ceiling.
- **Modest, not-yet-validated performance gain:** ROC-AUC of 0.6890 and PR-AUC of 0.6717 are only slightly higher than the Week 5 baseline (approximately +0.0017 ROC-AUC and +0.0037 PR-AUC). This small numerical gain has not yet been shown to be stable across patient-grouped cross-validation, and overall discrimination remains modest — a substantial share of both no-shows and non-no-shows are still misclassified.
- **Reliance on historical data:** the model is trained on past patterns. Changes in patient demographics, booking behaviour, clinic policy, or external shocks could cause data or concept drift, degrading performance if the model is not periodically retrained.
- **Threshold dependency and a provisional threshold:** classification performance is highly sensitive to the chosen probability threshold. A HealthConnect-aligned threshold (0.45) was derived using leakage-safe, out-of-fold evidence (Section 6.3), but it remains provisional because HealthConnect has not yet confirmed a formal tolerance for unnecessary patient contacts, and because the underlying cross-validation could not use patient-grouped separation (the processed feature matrices do not retain patient identifiers). Both points require Week 7 confirmation (Sections 12.1 and 12.2).

10.2 General Risks in Predictive Modelling for Healthcare

- **Algorithmic bias and fairness:** if historical data reflects systemic inequities (e.g. certain groups facing greater practical barriers to attendance), the model could perpetuate or amplify these patterns unless formally tested for disparate impact.
- **Patient experience and trust:** False Positives risk patient frustration or a sense of being unfairly monitored; interventions triggered by the model should be designed with this in mind.
- **Operational overheads:** intervention costs (extra reminders, outreach) scale with the False Positive rate, and could outweigh benefits if not properly managed.
- **Over-reliance and automation bias:** the model should augment, not replace, clinical and administrative judgement, particularly for edge cases it was not designed to recognise.
- **Data privacy and security:** patient data used for training and inference must continue to be handled in line with applicable privacy and healthcare-data obligations.

Proposed mitigations — continuous performance and drift monitoring, a defined retraining cadence, human-in-the-loop review of triggered interventions, regular ethical/fairness review, and transparent explanation of model outputs to stakeholders — are carried forward as explicit Week 7 testing requirements in Section 12.

11. Cross-track contribution

11.1 Integration Point

- **Track collaborated with:** Data Analytics Track. (The ML Engineering track was also identified as relevant for downstream pipeline integration of the candidate model, but the Week 6 integration activity itself was carried out with Data Analytics.)

- **Project dependency:** both tracks are investigating the same HealthConnect objective — reducing missed appointments — from complementary perspectives: Data Analytics identifies and validates patterns in appointment-attendance behaviour, while Data Science evaluates whether those patterns contribute to predicting future no-shows.
- **Information/output received from Data Analytics:** validated descriptive findings on appointment attendance and no-show patterns across patient characteristics, appointment characteristics, reminder patterns and operational/access segments (summarised in Section 11.2 below).
- **Information/output provided to Data Analytics:** the Week 6 error-analysis findings, the model's dominant predictive features, and the patient/appointment segments where the model produces relatively high False Positive or False Negative rates — enabling Data Analytics to target further investigation toward the segments where the model is weakest.

11.2 Data Analytics Findings Used in Week 6

- Overall baseline: after excluding Cancelled appointments, the eligible Attended/No-Show population had an overall no-show rate of 51.15%.
- Previous no-show history: one of the strongest relationships — 57.8% no-show rate for patients with at least one previous no-show, versus 46.3% with no prior no-show history.
- Booking lead time: a strong increasing pattern — 29.5% (0–7 days), 35.2% (8–14 days), 45.5% (15–30 days), 63.9% (31–60 days).
- Distance to clinic: a noticeable gradient — 48.7% (<5 km) rising to 60.5% (20+ km).
- Reminders: no reminder associated with a higher no-show rate (54.6%) than a reminder having been sent (49.9%).
- Waiting-time groups were found to be much more similar across categories and therefore less informative than lead time, previous history or distance.
- The Data Analytics track explicitly suggested testing interactions such as lead time × previous no-show history, lead time × reminder status, distance × lead time, and previous history × appointment type as candidate feature-refinement directions for Data Science.

11.3 What Changed as a Result

Two of the three Week 6 interaction features (history_x_lead_time and distance_x_lead_time; Section 3) were directly motivated by the Data Analytics track's suggested interaction directions. This is a concrete example of the cross-track exchange changing a modelling decision, rather than merely informing a narrative.

11.4 Mandatory Cross-Track Integration Evidence

The evidence table below records the concrete, verifiable artefact exchanged between the Data Science and Data Analytics tracks in Week 6, as required by the Week 6 brief:

Requirement	Week 6 evidence
Track collaborated with	Data Analytics
Project dependency	Data Science required analytical patterns/segments from Data Analytics to inform feature-refinement and error-

Requirement	Week 6 evidence
	analysis priorities.
Output received	Descriptive no-show findings for booking lead time, previous no-show history, distance to clinic, reminders, appointment type and related segments.
How the output was used	Findings were compared with baseline error-analysis results and used to justify targeted interaction hypotheses, in particular history $\times$ lead time and distance $\times$ lead time.
What changed because of integration	Feature refinement became targeted toward analytically supported relationships rather than arbitrary feature expansion; model interpretation was also checked against the Analytics observations.
Output provided by Data Science	Model-comparison results, False Positive/False Negative findings, coefficient/permutation-importance interpretation, and the provisional candidate model and threshold requirements for downstream use.
Evidence of actual exchange	https://analystlabfrica.slack.com/files/U0BK8QJUAUV/F0C1UUZQEC8/healthconnect_week5_dashboard.pdf

11.5 Validation of Cross-Track Findings Against Model Evidence

The Data Analytics track's descriptive findings were compared with the Refined Logistic Regression's coefficients and permutation importance. This comparison is supporting evidence, not causal validation: neither a model coefficient nor a permutation-importance value proves that a feature causes no-shows — it shows only that the model's learned associations are directionally consistent with Data Analytics' independently derived descriptive statistics.

Data Analytics Observation	Refined Logistic Regression Interpretation	Modelling Implication
Previous no-show history is strongly associated with higher no-show rates.	previous_no_shows has a positive coefficient (+0.293) and meaningful permutation importance (0.0211).	Supports retaining attendance history and testing history-based interactions.
No-show rates rise with booking lead time.	booking_lead_days has the largest positive coefficient (+0.599) and by far the highest permutation importance (0.1594).	Supports retaining lead time and testing history_x_lead_time / distance_x_lead_time.
Greater distance is associated with higher observed no-show rates.	distance_to_clinic_km has a positive coefficient (+0.166) and moderate permutation importance (0.0062).	Supports retaining distance and testing its interaction with lead time.
Reminder status/channel shows descriptive variation.	Reminder-related encoded features have smaller model contributions (reminder_sent_Yes: -0.141).	Treated as secondary evidence; reminder allocation is not assumed to cause attendance differences.

Data Analytics Observation	Refined Logistic Regression Interpretation	Modelling Implication
Age-group differences are comparatively small.	Age-related model effects are smaller/mixed relative to lead time and history.	Age is retained but not treated as a dominant standalone driver.
Follow-up appointments have a higher descriptive no-show rate.	The Follow-up indicator has a positive model association.	Appointment type remains relevant for interpretation and future segment testing.
Waiting-time groups are much more similar and less informative.	Not prominent among top coefficients or permutation importance.	Consistent with the Data Analytics finding of low informativeness; not prioritised for further feature work.

The strongest Data Analytics signals — particularly booking lead time and previous no-show history — are broadly consistent with the variables the Refined Logistic Regression relies on for prediction. This alignment across independent methods increases confidence in the Week 6 feature-refinement hypotheses, but, consistent with the caveat above, it does not establish causality and should not be over-interpreted as proof that the model has captured the true drivers of no-show behaviour.

12. Week 7 testing requirements

Week 7 should test whether the provisional Week 6 candidate model and its HealthConnect-aligned operating threshold remain reliable under stronger validation and are operationally appropriate. Six testing workstreams are recommended, each addressing a limitation or risk identified in Section 10:

12.1 Confirm the Provisional Operating Threshold with HealthConnect Requirements

Objective: validate whether the Week 6 threshold genuinely supports the aim of reducing missed appointments without degrading the patient support experience. Actions: use the Week 6 final_threshold_comparison results (Section 6.3) as the starting evidence rather than re-selecting a threshold from the holdout test set; confirm the acceptable number/proportion of unnecessary patient contacts with HealthConnect stakeholders; compare the additional no-shows identified against the additional false-positive interventions required; confirm whether the observed Extra FP per FN Avoided (approximately 1.14 on the holdout set) is operationally acceptable given reminder/outreach capacity and patient-experience expectations; and revise the threshold only through a new validation procedure if stakeholders establish a different acceptable trade-off.

12.2 Perform Robust Cross-Validation

Objective: determine whether the small numerical advantage of the Refined Logistic Regression and the selected operating threshold are stable across resamples. Actions: prefer StratifiedGroupKFold/GroupKFold using patient_id if patient identifiers can be recovered, since Week 5 deliberately used patient-aware separation and the Week 6 threshold-selection work (Section 6.3) could not; otherwise, clearly document the limitation of ordinary stratified cross-validation; report the mean and standard deviation of ROC-AUC, PR-AUC, Precision, Recall and F1 across folds; and compare the Refined Logistic Regression with the Week 5

baseline across folds to determine whether the observed +0.0017 ROC-AUC / +0.0037 PR-AUC advantage is consistent rather than sample-specific.

12.3 Test Model Fairness and Error Rates Across Patient Groups

Objective: check whether model errors are disproportionately concentrated in relevant demographic or access groups. Actions: compare Recall, Precision, false-positive rate and false-negative rate across available groups such as gender and age group, where appropriate; and investigate material disparities before operational use and document any mitigation required.

12.4 Validate Operational Impact

Objective: determine whether the selected threshold is practical for HealthConnect workflows. Actions: translate the threshold into the expected number/proportion of patients flagged for intervention; compare intervention capacity with the number of additional no-shows identified and additional false-positive contacts generated; define what action follows a high-risk prediction (for example, an additional reminder or staff outreach); and ensure the intervention remains proportionate and supportive rather than burdensome for patients.

12.5 Develop a Production Monitoring Plan

Track model discrimination, Precision/Recall, calibration where relevant, and realised False Positive/False Negative rates after deployment; monitor important feature distributions such as booking lead time, previous no-shows and distance for drift; monitor intervention volume and patient-support outcomes associated with the chosen operating threshold; and define alert thresholds and retraining/review criteria before production use.

12.6 Document Deployment and Integration Requirements

Define the exact inference-time feature schema and pre-processing steps; save the model together with the feature order/engineering logic required to reproduce predictions; specify model outputs (no-show probability plus threshold-based risk flag), expected inference latency and downstream HealthConnect integration points; and provide the candidate-model requirements and artefacts to the relevant downstream/ML Engineering track as part of the Week 7 hand-off.

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